$\Rightarrow R(T) = V$

Hence T is onto.

(e) $\Rightarrow$ (f)

Let T is onto

$\Rightarrow R(T) = V$

$\Rightarrow \dim(R(T)) = \dim(V)$

$\Rightarrow r(T) = p$.

(f) $\Rightarrow$ (g)

By the rank-nullity theorem,

$\dim(U) = r(T) + n(T)$

$\Rightarrow p = p + n(T)$

$\Rightarrow n(T) = 0$.

(g) $\Rightarrow$ (h)

Let $n(T) = 0$

$\Rightarrow N(T) = \{0_v\}$

$\Rightarrow$ T is one-one

Again, $\dim(U) = \dim(V)$

So T is onto.

Hence T^{-1} exists.

(h) $\Rightarrow$ (a)

Let T^{-1} exists.

$\Rightarrow$ T must be one-one and onto.

$\therefore$ T is non-singular.

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